Prairie Sun — 50 MWdc solar farm

Texas (statewide factor)

TOTAL PROJECT COST	PER SF	PER MW	HARD / SOFT / LAND
$76,697,263	**—**	**$1,533,945**	**84 / 13 / 3%**

COST SUMMARY

Construction hard cost	**$59,891,774**
FF&E	**$0**
Design + construction contingency	**$4,881,180**
Hard cost subtotal	**$64,772,953**
Soft costs	**$4,252,112**
Impact & tap fees	**$0**
Developer fee	**$2,777,854**
Financing (interest + fees)	**$2,145,252**
Soft cost contingency	**$421,292**
Soft cost subtotal	**$9,596,510**
Land & acquisition	**$2,327,800**
Total project cost	**$76,697,263**

BASIS

Location	**Texas (factor 0.820)**
Gross building SF	**—**
Site acres	**250.00**
Cost per acre	**$306,789**
Research data version	**2026.0c737c8255**
Engine	**budget-engine-2.0.0**

All defaults come from the research data shipped with the app (2026.0c737c8255), escalated to the construction midpoint and adjusted to the project location. Values marked "Derived estimate — verify" have no direct published source.

Cost by Component

COMPONENT	TYPE	SIZE	HARD (INCL. FF&E, CONT.)	SOFT (INCL. FIN.)	LAND	TOTAL	PER PRIMARY UNIT	PER SF	RANGE CHECK
Solar array	Solar farm (utility / community)	50 MW	$64,772,953	$9,596,510	$2,327,800	$76,697,263	$1,533,945	—	within ($925,284–$2,177,139)
TOTAL			**$64,772,953**	**$9,596,510**	**$2,327,800**	**$76,697,263**			

Cost by Phase

PHASE	START	MONTHS	MIDPOINT	ESCALATION	HARD	SOFT	LAND	TOTAL	LOAN	INTEREST
Phase 1	2027-03	19	2028-01	6.3%	$64,772,953	$9,596,510	$2,327,800	$76,697,263	$46,018,358	$1,685,068
TOTAL					**$64,772,953**	**$9,596,510**	**$2,327,800**	**$76,697,263**		

Line-Item Detail

COMPONENT	PHASE	GROUP	CODE	LINE	QTY	UNIT	RATE	AMOUNT	EDITED
Solar array	Phase 1	Construction	01 00 00	General Requirements	—	—	—	$6,991,622	—
Solar array	Phase 1	Construction	03 00 00	Concrete	—	—	—	$443,260	—
Solar array	Phase 1	Construction	25 00 00	Integrated Automation	—	—	—	$196,497	—
Solar array	Phase 1	Construction	26 00 00	Electrical	—	—	—	$5,108,911	—
Solar array	Phase 1	Construction	31 00 00	Earthwork	—	—	—	$1,476,009	—
Solar array	Phase 1	Construction	32 00 00	Exterior Improvements	—	—	—	$1,032,749	—
Solar array	Phase 1	Construction	48 00 00	Electrical Power Generation — balance	—	—	—	$13,696,233	—
Solar array	Phase 1	Construction	48 00 00	PV module supply (bifacial mono) (within baseline)	50	MW	$364,505.67	$18,225,283	—
Solar array	Phase 1	Construction	33 00 00	Project substation & main power transformer (within baseline)	1	LS	$10,344,079.75	$10,344,080	—
Solar array	Phase 1	Specialty & site	33 00 00	Generator tie / transmission line to POI	5,280	LF	$450.21	$2,377,130	—
Solar array	Phase 1	Contingency	99 02 00	Design contingency	3	% of hard	—	$1,796,753	—
Solar array	Phase 1	Contingency	99 03 00	Hard cost (construction) contingency	5	% of hard	—	$3,084,426	—
Solar array	Phase 1	Soft costs	SC	Architecture & engineering (basic services, design + CA)	3	% of hard	—	$2,033,639	—

COMPONENT	PHASE	GROUP	CODE	LINE	QTY	UNIT	RATE	AMOUNT	EDITED
Solar array	Phase 1	Soft costs	SC	Civil engineering, ALTA/topo survey, geotechnical report	1	% of hard	—	$312,867	—
Solar array	Phase 1	Soft costs	SC	Building permit & plan review fees (excl. impact / tap fees)	1	% of hard	—	$312,867	—
Solar array	Phase 1	Soft costs	SC	Legal, accounting, cost certification	0	% of hard	—	$219,007	—
Solar array	Phase 1	Soft costs	SC	Builder's risk insurance	0	% of hard	—	$125,147	—
Solar array	Phase 1	Soft costs	SC	Owner's general liability / OCIP-OCP during construction	0	% of hard	—	$93,860	—
Solar array	Phase 1	Soft costs	SC	Marketing, lease-up / rent-up (excl. commercial leasing commissions)	0	% of hard	—	$250,294	—
Solar array	Phase 1	Soft costs	SC	Construction-loan title, closing & recording	0	% of hard	—	$125,147	—
Solar array	Phase 1	Soft costs	SC	Special inspections, materials testing, commissioning	0	% of hard	—	$250,294	—
Solar array	Phase 1	Soft costs	SC	Owner's representative / project management	1	% of hard	—	$469,301	—
Solar array	Phase 1	Soft costs	SC	Third-party reports (appraisal, Phase I ESA, market study, PCA/CNA)	1	project	$20,500.00	$20,500	—
Solar array	Phase 1	Contingency	SC	Soft cost contingency	10	% of soft	—	$421,292	—
Solar array	Phase 1	Land	LAND	Land / acquisition	—	—	—	$2,250,000	—
Solar array	Phase 1	Land	LAND	Land closing costs	—	—	—	$45,000	—
Solar array	Phase 1	Land	LAND	Land due diligence	—	—	—	$32,800	—
Solar array	Phase 1	Soft costs	SC	Real estate taxes during construction	2	years	$24,750.00	$39,188	—
Solar array	Phase 1	Developer fee	DEV	Developer fee	4	% of cost excl. land	—	$2,777,854	—
Solar array	Phase 1	Financing	FIN	Construction loan interest (capitalized)	—	—	—	$1,685,068	—
Solar array	Phase 1	Financing	FIN	Construction loan fee	—	—	—	$460,184	—
TOTAL				**Total project cost**				**$76,697,263**	

Draw Schedules

PHASE 1 — MONTHLY DRAWS

MONTH	HARD	SOFT	LAND	EQUITY	LOAN DRAW	INTEREST	LOAN BALANCE
2027-03	$0	$2,460,866	$2,327,800	$5,248,850	$0	$0	$0
2027-04	$441,709	$262,652	$0	$704,362	$0	$0	$0
2027-05	$1,313,079	$262,652	$0	$1,575,732	$0	$0	$0
2027-06	$2,148,632	$262,652	$0	$2,411,284	$0	$0	$0
2027-07	$2,925,575	$262,652	$0	$3,188,228	$0	$0	$0
2027-08	$3,622,717	$262,652	$0	$3,885,369	$0	$0	$0
2027-09	$4,221,040	$262,652	$0	$4,483,692	$0	$0	$0
2027-10	$4,704,224	$262,652	$0	$4,966,876	$0	$0	$0
2027-11	$5,059,089	$262,652	$0	$4,214,513	$1,107,229	$0	$1,107,229
2027-12	$5,275,956	$262,652	$0	$0	$5,538,608	$6,459	$6,652,296
2028-01	$5,348,908	$262,652	$0	$0	$5,611,560	$38,805	$12,302,661
2028-02	$5,275,956	$262,652	$0	$0	$5,538,608	$71,766	$17,913,035
2028-03	$5,059,089	$262,652	$0	$0	$5,321,742	$104,493	$23,339,269
2028-04	$4,704,224	$262,652	$0	$0	$4,966,876	$136,146	$28,442,291
2028-05	$4,221,040	$262,652	$0	$0	$4,483,692	$165,913	$33,091,897
2028-06	$3,622,717	$262,652	$0	$0	$3,885,369	$193,036	$37,170,302
2028-07	$2,925,575	$262,652	$0	$0	$3,188,228	$216,827	$40,575,357
2028-08	$2,148,632	$262,652	$0	$0	$2,411,284	$236,690	$43,223,330
2028-09	$1,313,079	$262,652	$0	$0	$1,575,731	$252,136	$45,051,198
2028-10	$441,709	$262,652	$0	$0	$704,362	$262,799	$46,018,358

Sources & Assumptions

ASSUMPTIONS

Solar array — baseline price	**$1,142,998 / MW · Researched (agriculture-energy/solar-farm.json)**
Solar array — factors	**regional 0.82 · quality 1 · size 1.057 · escalation 1.0626**
Solar array — research note	**Ground-mount single-axis-tracking PV priced per MWdc. Cost is dominated by modules, trackers/racking and electrical BOS. Scale economies are strong: LBNL finds projects over 250 MWac cost about 60% less per watt than projects under 20 MWac, so community-scale (1–20 MWac) projects sit at the high end. Data quality is high: an itemized Sargent & Lundy/EIA estimate, the DOE/NREL bottom-up benchmark and LBNL empirical plant data all agree within about 15%.**
Phase 1 — schedule	**2027-03, 19 months, midpoint 2028-01, escalation 6.26% (forecast rates from _indices/escalation.json)**
Phase 1 — financing	**LTC 60%, rate 7.00%, fee 1.00%**

SOURCES (35)

REF	SOURCE	PUBLISHER	VALUE	CONFIDENCE	URL
_common/financing.json#e1	Secured Overnight Financing Rate (SOFR)	Federal Reserve Bank of St. Louis (FRED)	0.039 pct (2026)	high	https://fred.stlouisfed.org/series/SOFR
_common/financing.json#e2	30-Day Average SOFR (SOFR30DAYAVG)	Federal Reserve Bank of St. Louis (FRED)	0.03889 pct (2026)	high	https://fred.stlouisfed.org/series/SOFR30DAYAVG
_common/financing.json#e3	Bank Prime Loan Rate (DPRIME)	Federal Reserve Bank of St. Louis (FRED)	0.07 pct (2026)	high	https://fred.stlouisfed.org/series/DPRIME
_common/financing.json#e4	Federal Funds Target Range - Upper Limit (DFEDTARU)	Federal Reserve Bank of St. Louis (FRED)	0.0375–0.04 pct (2026)	high	https://fred.stlouisfed.org/series/DFEDTARU
_common/financing.json#e5	12 CFR Appendix A to Subpart A of Part 365 - Interagency Guidelines for Real Estate Lending Policies	Legal Information Institute, Cornell Law School (text of FDIC regulation)	0.8 pct	high	https://www.law.cornell.edu/cfr/text/12/appendix-A_to_subpart_A_of_part_365
_common/financing.json#e6	The July 2026 Senior Loan Officer Opinion Survey on Bank Lending Practices	Board of Governors of the Federal Reserve System	text (2026)	high	https://www.federalreserve.gov/data/sloos/sloos-202607.htm
_common/land-closing.json#e1	Title Rate Chart (effective March 1, 2026)	Texas Department of Insurance	5226 USD/policy (2026)	high	https://www.tdi.texas.gov/title/titlerates2026.html
_common/land-closing.json#e2	Real estate excise tax	Washington State Department of Revenue	0.0128 pct (2026)	high	https://dor.wa.gov/taxes-rates/other-taxes/real-estate-excise-tax
_common/land-closing.json#e3	Realty Transfer Tax	Pennsylvania Department of Revenue	0.01 pct (2026)	high	https://www.pa.gov/en/agencies/revenue/resources/tax-types-and-information/realty-transfer-tax.html
_common/land-closing.json#e4	Documentary Stamp Tax	Florida Department of Revenue	0.007 pct (2026)	high	https://floridarevenue.com/taxes/taxesfees/Pages/doc_stamp.aspx
_common/land-closing.json#e5	PHFA 2025 Underwriting Application — Development Cost Limits (revised 1-17-25)	Pennsylvania Housing Finance Agency	text (2025)	high	https://www.phfa.org/forms/multifamily_application_guidelines/application/2025-2026-mai-07.pdf
_common/soft-costs.json#e1	Chapter 08: Architect/Engineer Fee Guidelines and Fee Schedule (2027-29 capital budget instructions)	Washington State Office of Financial Management	0.0852 pct (2026)	high	https://ofm.wa.gov/budget/state-budget-2027-29/instructions/capital/chapter-08/
_common/soft-costs.json#e10	Builders Risk Insurance in 2026: What It Costs, Who Pays, and the Exclusions That Wreck Commercial Projects	Terrapin Construction Group	0.0035–0.009 pct (2026)	low	https://terrapincg.com/news/builders-risk-insurance-commercial-construction-cost-2026
_common/soft-costs.json#e11	Builders Risk Insurance in 2026: What It Costs, Who Pays, and the Exclusions That Wreck Commercial Projects	Terrapin Construction Group	0.0007–0.0019 pct (2026)	low	https://terrapincg.com/news/builders-risk-insurance-commercial-construction-cost-2026
_common/soft-costs.json#e13	PHFA 2025 Underwriting Application — Development Cost Limits (revised 1-17-25)	Pennsylvania Housing Finance Agency	text (2025)	medium	https://www.phfa.org/forms/multifamily_application_guidelines/application/2025-2026-mai-07.pdf
_common/soft-costs.json#e14	What are Typical Developer Fees on a Multifamily Development?	HelloData	0.03–0.05 pct (2025)	low	https://www.hellodata.ai/help-articles/what-are-typical-developer-fees-on-a-multifamily-development

REF	SOURCE	PUBLISHER	VALUE	CONFIDENCE	URL
_common/soft-costs.json#e15	2025-2026 Delaware State Housing Authority Low Income Housing Tax Credit Guidelines	Delaware State Housing Authority	0.15 pct (2025)	high	https://www.destatehousing.com/wp-content/uploads/2026/06/2025-2026-LIHTC-Guidelines.pdf
_common/soft-costs.json#e16	PHFA 2025 Underwriting Application — Development Cost Limits (revised 1-17-25)	Pennsylvania Housing Finance Agency	0.15 pct (2025)	high	https://www.phfa.org/forms/multifamily_application_guidelines/application/2025-2026-mai-07.pdf
_common/soft-costs.json#e17	PHFA 2025 Underwriting Application — Development Cost Limits (revised 1-17-25)	Pennsylvania Housing Finance Agency	1200 USD/unit (2025)	high	https://www.phfa.org/forms/multifamily_application_guidelines/application/2025-2026-mai-07.pdf
_common/soft-costs.json#e18	2025-2026 Delaware State Housing Authority Low Income Housing Tax Credit Guidelines	Delaware State Housing Authority	500 USD/unit (2025)	high	https://www.destatehousing.com/wp-content/uploads/2026/06/2025-2026-LIHTC-Guidelines.pdf
_common/soft-costs.json#e19	Title Rate Chart (effective March 1, 2026)	Texas Department of Insurance	0.005226 pct (2026)	high	https://www.tdi.texas.gov/title/titlerates2026.html
_common/soft-costs.json#e2	Chapter 08: Architect/Engineer Fee Guidelines and Fee Schedule (2027-29 capital budget instructions)	Washington State Office of Financial Management	0.0988 pct (2026)	high	https://ofm.wa.gov/budget/state-budget-2027-29/instructions/capital/chapter-08/
_common/soft-costs.json#e20	Documentary Stamp Tax	Florida Department of Revenue	0.0035 pct (2026)	high	https://floridarevenue.com/taxes/taxesfees/Pages/doc_stamp.aspx
_common/soft-costs.json#e21	Board of Commissioners agenda item — Edneyville Elementary School proposal (Feb 15, 2017)	Henderson County, North Carolina	0.01 pct (2017)	high	https://www.hendersoncountync.gov/sites/default/files/archives/Meetings/Board_of_Commissioners/agenda170215/DI.B.pdf
_common/soft-costs.json#e3	Chapter 08: Architect/Engineer Fee Guidelines and Fee Schedule (2027-29 capital budget instructions)	Washington State Office of Financial Management	0.0716 pct (2026)	high	https://ofm.wa.gov/budget/state-budget-2027-29/instructions/capital/chapter-08/
_common/soft-costs.json#e4	PHFA 2025 Underwriting Application — Development Cost Limits (revised 1-17-25)	Pennsylvania Housing Finance Agency	0.0483 pct (2025)	high	https://www.phfa.org/forms/multifamily_application_guidelines/application/2025-2026-mai-07.pdf
_common/soft-costs.json#e5	Board of Commissioners agenda item — Edneyville Elementary School proposal (Feb 15, 2017)	Henderson County, North Carolina	0.06 pct (2017)	high	https://www.hendersoncountync.gov/sites/default/files/archives/Meetings/Board_of_Commissioners/agenda170215/DI.B.pdf
_common/soft-costs.json#e6	Board of Commissioners agenda item — Edneyville Elementary School proposal (Feb 15, 2017)	Henderson County, North Carolina	0.12 pct (2017)	high	https://www.hendersoncountync.gov/sites/default/files/archives/Meetings/Board_of_Commissioners/agenda170215/DI.B.pdf
_common/soft-costs.json#e7	Building Valuation Data – August 2025	International Code Council (posted by City of Kirkland, WA)	0.0075 pct (2025)	low	https://www.kirklandwa.gov/files/sharedassets/public/v/2/development-services/pdfs/building-pdfs/icc-august-2025.pdf
_common/soft-costs.json#e8	PHFA 2025 Underwriting Application — Development Cost Limits (revised 1-17-25)	Pennsylvania Housing Finance Agency	175000 USD/project (2025)	high	https://www.phfa.org/forms/multifamily_application_guidelines/application/2025-2026-mai-07.pdf
_common/soft-costs.json#e9	2025-2026 Delaware State Housing Authority Low Income Housing Tax Credit Guidelines	Delaware State Housing Authority	150000 USD/project (2025)	high	https://www.destatehousing.com/wp-content/uploads/2026/06/2025-2026-LIHTC-Guidelines.pdf

REF	SOURCE	PUBLISHER	VALUE	CONFIDENCE	URL
agriculture-energy/solar-farm.json#e1	Capital Cost and Performance Characteristic Estimates for Utility Scale Electric Power Generating Technologies (Sargent & Lundy for EIA), Case 16 Solar PV with Single-Axis Tracking, 150 MWAC	U.S. Energy Information Administration / Sargent & Lundy	1082487 USD/mw (2023)	high	https://www.eia.gov/analysis/studies/powerplants/capitalcost/pdf/capital_cost_AEO2025.pdf
agriculture-energy/solar-farm.json#e2	Utility-Scale Solar, 2024 Edition — Executive Summary	Lawrence Berkeley National Laboratory	1080000 USD/mw (2023)	high	https://eta-publications.lbl.gov/sites/default/files/2024-10/utility-scale_solar_2024_executive_summary.pdf
agriculture-energy/solar-farm.json#e3	Solar Photovoltaic System Cost Benchmarks (2025Q1 benchmarks)	U.S. Department of Energy, Solar Energy Technologies Office (NREL modeling)	1120000 USD/mw (2024)	high	https://www.energy.gov/eere/solar/solar-photovoltaic-system-cost-benchmarks
agriculture-energy/solar-farm.json#e4	Cost of Utility-Scale Solar: LCOE, PPAs, and Storage (summarizing LBNL U.S. Utility-Scale Solar, 2025 Data Update slides)	LegalClarity (secondary, citing Lawrence Berkeley National Laboratory)	1220000 USD/mw (2024)	medium	https://legalclarity.org/cost-of-utility-scale-solar-lcoe-ppas-and-storage/

Sample output — revenue, land and exit-cap inputs are illustrative. Engine budget-engine-2.0.0.